EXPERIMENT 14: CAPTURE, SAVE AND ANALYZE NETWORK TRAFFIC USING WIRESHARK

AIM:

To capture, save and analyze network traffic involving protocols such as TCP, UDP, IP, HTTP, ARP, DHCP, ICMP and DNS using Wireshark.

PROCEDURE:

1. Open Wireshark and select the active network interface.

2. Start capturing traffic and generate permitted test traffic using ping or a browser.

3. Use display filters to isolate protocols of interest.

4. Select packets and inspect the Ethernet, IP, transport and application protocol fields.

5. Stop the capture and save it as a .pcapng file.

6. Open the saved capture and review packet details and protocol statistics.

7. Select the interface that carries the traffic to be studied and confirm that packet counters are increasing.
8. Use one display filter at a time and clear the filter to return to the full capture.
9. Inspect source and destination addresses, ports, flags and protocol details in selected packets.
10. Save the capture with a meaningful filename and reopen it to confirm that it was saved correctly.

CODE:

```
# Generate sample traffic in a terminal:
ping -c 4 8.8.8.8
nslookup example.com

# Useful Wireshark display filters:
tcp
```

```
udp
ip
arp
icmp
dns
dhcp
http

# Start a terminal capture (Linux; interface may vary):
sudo tcpdump -i any -w lab_traffic.pcap

# Open the saved capture:
wireshark lab_traffic.pcap
```

OUTPUT:



RESULT:

Thus, network traffic was captured, saved and analyzed using Wireshark display filters and protocol-layer details.